

HR Candidate Ranking using Group VIKOR

Introduction

This project presents a decision support system for ranking HR candidates using the Group VIKOR multi-criteria decision-making method in Python. The objective is to evaluate and prioritize candidates based on multiple HR-related criteria derived from a real-world HR analytics dataset.

The project combines data preprocessing, criteria weighting, candidate ranking and visualization techniques in order to support intelligent recruitment analysis.

Dataset

The analysis uses an HR dataset obtained from Kaggle containing candidate information such as:

- Education level
- Professional experience
- Company size
- Training hours
- City development index

Categorical variables were transformed into numerical scores to enable quantitative analysis.

Methodology

The project follows these main steps:

1. Data preprocessing and cleaning
2. Transformation of categorical variables into numerical scores
3. Definition of evaluation criteria
4. Assignment of weights from multiple decision makers
5. Aggregation of group weights
6. Application of the Group VIKOR method
7. Ranking of candidates based on Q, S and R values
8. Visualization and export of results to Excel

The decision makers included:

- HR Manager
- Technical Lead
- CEO

Technologies Used

- Python
- Pandas

- NumPy
- Matplotlib
- XlsxWriter

Results

The system generates:

- Candidate rankings
- Group utility scores
- VIKOR Q/S/R values
- Weight analysis charts
- Excel reports with visualizations

The final output helps identify the most suitable candidates according to the defined multi-criteria evaluation process.

Conclusion

This project demonstrates how multi-criteria decision-making techniques can be applied to HR analytics and recruitment support systems. Group VIKOR provides a structured and explainable approach for candidate evaluation using multiple decision makers and quantitative criteria.

References

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